

RegView

AI-Powered Regulatory Search Platform

Architecture & Implementation Overview

Version	v1.0
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Audience	Engineering · Product · Stakeholders

1. Overview

RegView answers plain-English questions about FDA drugs, patents, clinical trials, and recalls. It ingests public regulatory data (openFDA, ClinicalTrials.gov, FDA Orange Book) plus any PDFs or URLs you provide, stores everything as searchable vectors in a local database (ChromaDB), retrieves the most relevant snippets for each question, and asks Anthropic's Claude to write a short, cited answer.

If nothing relevant is found in the local library, Claude falls back to its own general knowledge and clearly labels the answer. This makes the system safe by default: users always know whether an answer came from the regulatory library or from the model's training data.

Design goals

- **Local-first.** Vectors, documents, and session history live on disk — no third-party data store required.
- **Auditable.** Every grounded answer cites the exact chunks it used.
- **Cost-controlled.** Retrieval trims LLM context to only relevant snippets; Haiku is the default model.
- **Swappable.** Vector DB, LLM, and session store are all behind small interfaces.

2. System Architecture

Four layers, top to bottom: **Client**, **Gateway**, **Services**, and **Storage / External APIs**. Solid arrows show the live request path; dashed arrows show data access.

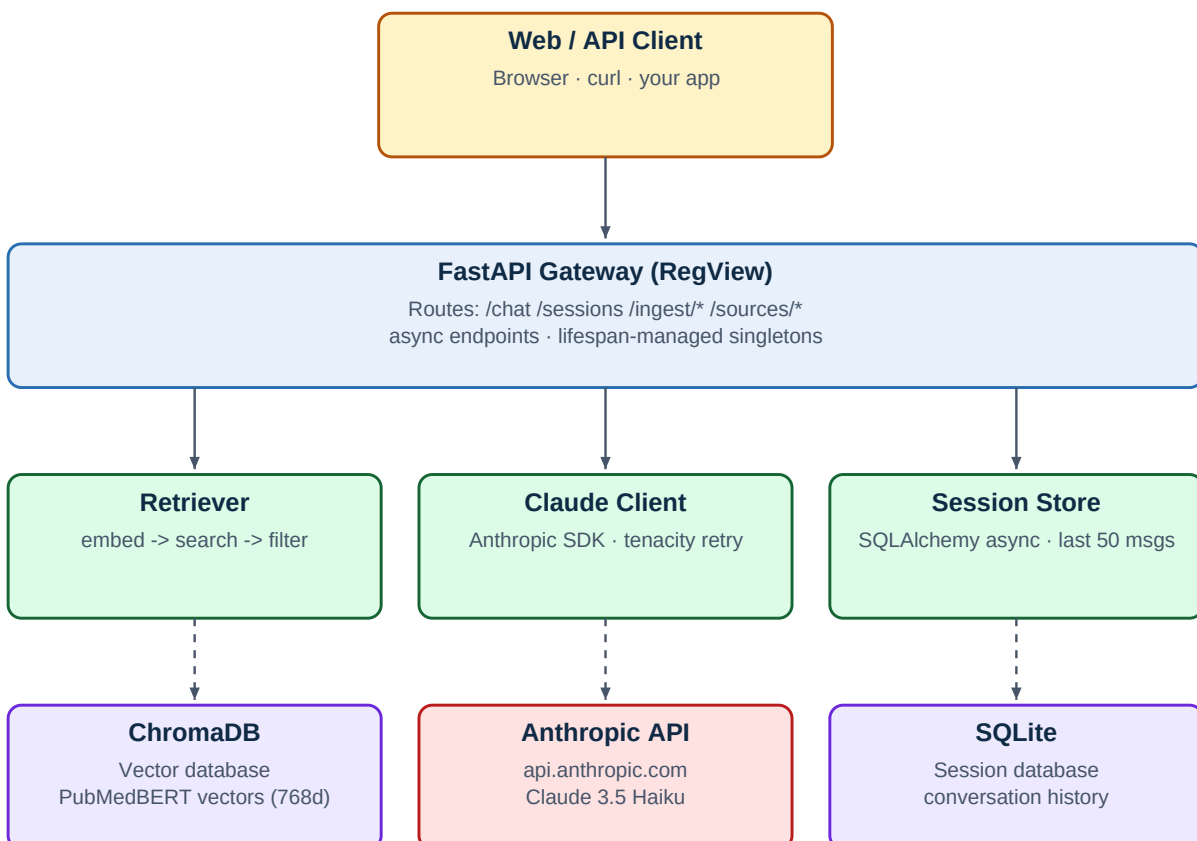


Figure 1 — System architecture (four-layer view)

3. Components

Layer	Responsibility
FastAPI gateway	HTTP endpoints; wires everything together at startup via lifespan-managed singletons.
Retriever	Embeds the question, queries the vector store, filters by distance, returns top-K chunks.
Embedding model	PubMedBERT (biomedical BERT), loaded once and cached. 768-dim vectors.
Vector store	ChromaDB with cosine distance and metadata filters.
Claude client	Anthropic SDK wrapper with tenacity retries and structured response parsing.
Session store	SQLAlchemy 2.x async persists the last N messages per session.
Ingestion pipeline	Chunker (800 chars / 120 overlap) then embedder then vector store. Deduplicates by content hash.
Connectors	One module per source: openFDA labels, FAERS, 510(k), enforcement, ClinicalTrials.gov, Orange Book.

4. Chat Request Flow

The end-to-end path taken by every user question. Steps 3 and 6 form a two-tier relevance filter (mechanical distance cutoff, then Claude's semantic check) that keeps off-topic content out of the final answer.

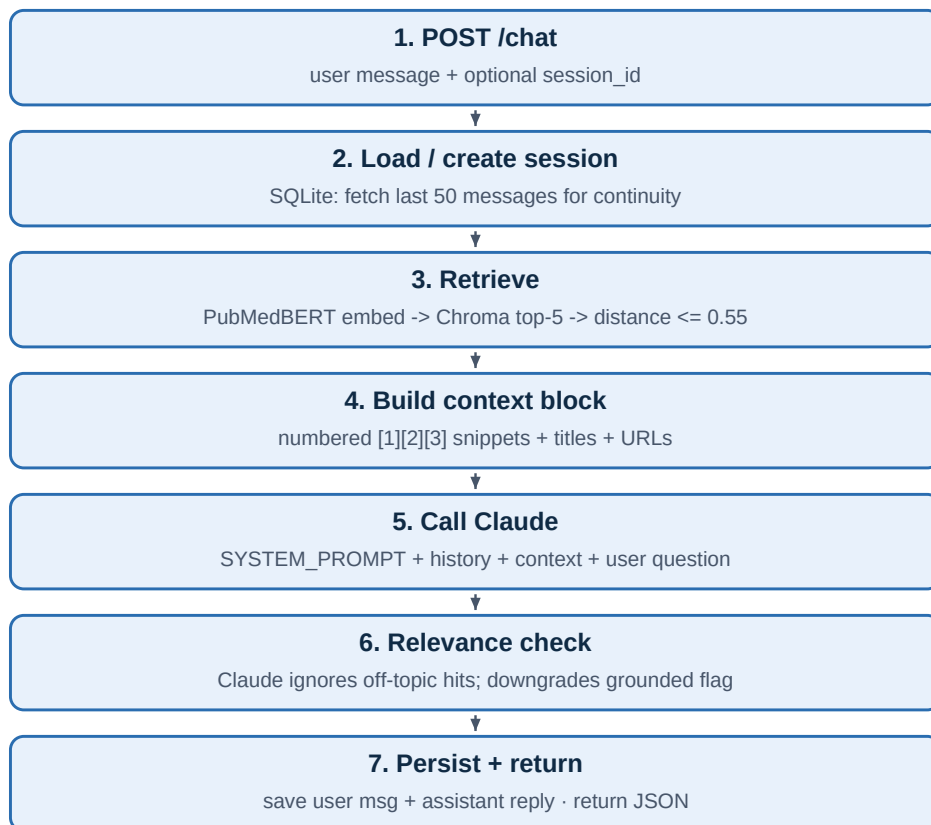


Figure 2 — /chat request flow (7 steps)

5. Ingestion Pipeline

Every source — user-provided PDFs, URLs, or bulk feeds from openFDA and ClinicalTrials.gov — flows through the same five stages. Chunk and document IDs are deterministic content hashes, so re-ingesting is safe: existing content is skipped and only new material is added.



Figure 3 — Ingestion pipeline (five stages)

6. Relevance Filtering

Tier 1 — Distance threshold (mechanical). Every retrieved chunk gets a cosine distance score. Chunks with `distance > RAG_DISTANCE_THRESHOLD` are dropped before Claude ever sees them.

Tier 2 — Claude relevance check (semantic). The system prompt instructs Claude to judge each surviving snippet on its own merits and silently ignore topically-irrelevant hits (for example, a drug label mentioning 'PM' for post-meridien dosing when the user asked about a Prime Minister). If every snippet fails, Claude uses the fallback disclaimer and the API downgrades the response to `grounded: false, citations: []`.

Distance intuition (cosine)

Distance	Meaning	What happens
0.00 – 0.15	Near-identical	Cited with high confidence
0.15 – 0.30	Very relevant	Cited
0.30 – 0.55	Loosely related	Sent to Claude; Claude may still ignore
> 0.55 (default cutoff)	Unrelated	Dropped by retriever
1.0 – 2.0	Opposite meaning	Never seen — filtered long before

7. Local-First, Claude-Fallback Logic

Every question tries the local library first. The final behaviour depends on two questions: did any chunks pass the distance threshold, and (when Claude is configured) did Claude judge them relevant?

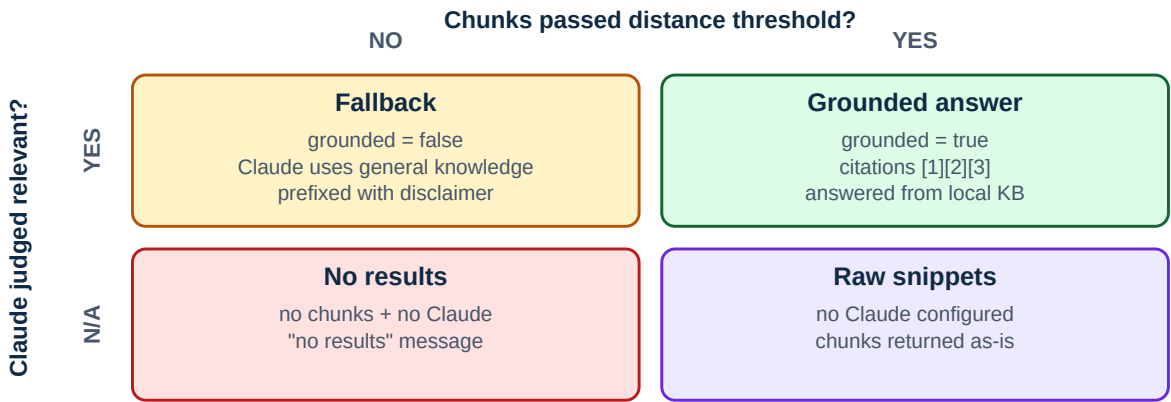


Figure 4 — Local-first / Claude-fallback decision matrix

8. Technology Stack

Concern	Technology	Why chosen
Web framework	FastAPI + Uvicorn	Async, auto-generated OpenAPI documentation
LLM	Anthropic Claude (Haiku / Sonnet / Opus)	Strong reasoning + long context; SDK with streaming support
Embeddings	PubMedBERT via sentence-transformers	Biomedical vectors outperform generic BERT on FDA text
Vector DB	ChromaDB	Cosine distance, metadata filters, embeddable
Session DB	SQLite via SQLAlchemy 2.x async	Lightweight persistence for conversation history
HTTP client	httpx (async)	For openFDA / ClinicalTrials.gov fetches
Retries	tenacity	Exponential backoff around Claude and openFDA
Config	pydantic-settings	Type-checked, environment-driven configuration
Logging	loguru	Structured, colour-coded application logs

9. Data Sources

Source	Endpoint	Volume (bulk)
openFDA drug labels	api.fda.gov/drug/label.json	Up to 25,000 (API cap)
openFDA FAERS (adverse events)	api.fda.gov/drug/event.json	Per-drug aggregated summary
openFDA 510(k) devices	api.fda.gov/device/510k.json	Up to 25,000
openFDA enforcement / recalls	api.fda.gov/{drug,device,food}/enforcement.json	Up to 25,000 per kind
FDA Orange Book	Public FDA archive (products, patents, exclusivity)	~34,000 products
ClinicalTrials.gov v2	clinicaltrials.gov/api/v2/studies	Unlimited (paginated)
User documents	PDF / DOCX uploads and seed URLs	Ad-hoc

10. Out of Scope (by design)

- **No Model Context Protocol (MCP) server.** Direct SDK calls are simpler and faster for a fixed RAG pipeline.
- **No authentication or rate-limiting** at the API layer. A gateway is expected in front for production use.
- **No streaming responses yet.** Non-streaming responses keep the code simple; streaming can be added later.
- **No FAERS bulk ingest.** Millions of raw events are not useful as vectors; we aggregate per drug instead.
- **No hosted vector DB.** ChromaDB covers current volumes; larger corpora can move to Qdrant or Pinecone.

Disclaimer. RegView is for research and informational use only. It is not medical, legal, or regulatory advice. Always verify important findings against the primary sources (FDA.gov, ClinicalTrials.gov, FDA Orange Book).

11. Glossary

Term	Meaning
RAG	Retrieval-Augmented Generation. The pattern of retrieving relevant snippets from a private library and passing them to an LLM as extra context.
Embedding	A high-dimensional numeric vector representing the meaning of a piece of text. Similar meanings sit close together in vector space.
Cosine distance	Distance metric used by ChromaDB: 0.0 means identical direction, 1.0 means orthogonal, 2.0 means opposite. Lower is better.
Chunk	A ~800-character slice of a source document. Each chunk is embedded and stored as an independent search unit.
Grounded answer	An answer supported by citations to retrieved chunks. Marked grounded: <code>true</code> in the API response.
Fallback answer	An answer produced from Claude's general knowledge because the local library had no relevant material. Always prefixed with a disclaimer.
openFDA	The FDA's public JSON API for drugs, devices, foods, and recalls. See open.fda.gov .